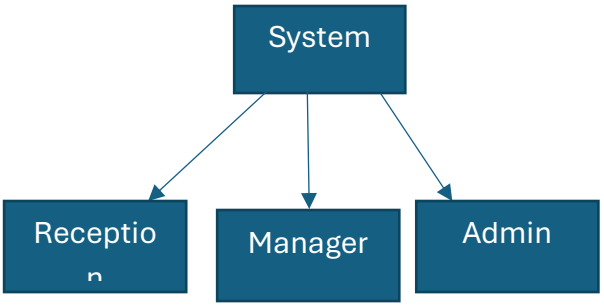
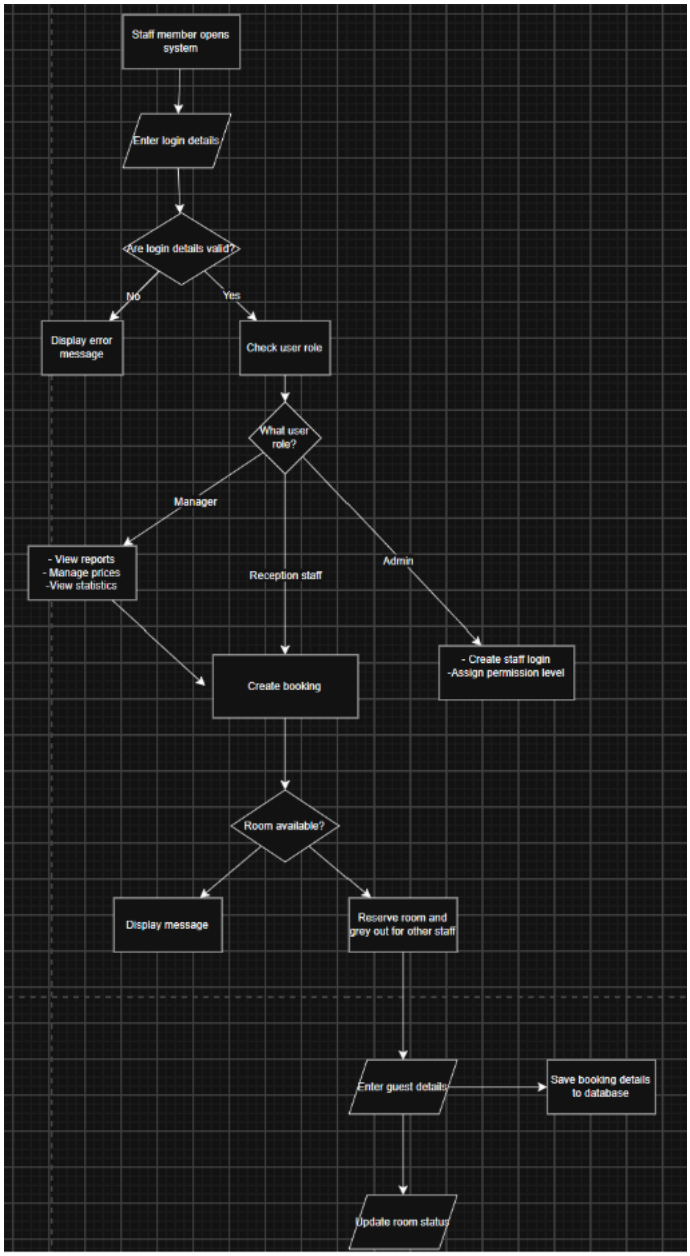
Hotel Management System NEA Design

Overall system:

- The system will be based divided into sections based on whether user is receptionist, manager or admin
- Each role will perform different tasks but some will be able to do the same tasks




Staff member opens system
Enter login details
Are login details valid?
No
Yes
Display error message
Check user role
What user role?
Manager
Reception staff
Admin
- View reports
- Manage prices
-View statistics
Create booking
- Create staff login
-Assign permission level
Room available?
Display message
Reserve room and grey out for other staff
Enter guest details
Save booking details to database
Update room status

Description of algorithms:

Booking algorithms:

1. Booking request is recieved
2. Staff will discuss details of booking with guest such as check in date, type of room
3. When a staff has selected a room it will be instantly greyed out for other staff so the same room cannot be booked
4. Guest details are entered and stored in the guest database where the booking will be assigned to the guest
5. Booking details will be entered and will be stored in the booking database
6. Prices will be added to the bookings database

Additional services algorithms:

1. The guest is checked in
2. Guest chooses their additional service
3. Guest pays the price of that additional services
4. The cost is added to the variable “price” assigned to the guest

Login system:

Users enter their username and password (password will be dotted for security purposes). The system will search the Users database for credentials that match with the credentials that user entered. If the credentials match then user will be taken to the next screen.



If credentials do not match the user will be shown an error message and the credentials entered will be reset.



Guest management:

All users will have access to the guest management screen after logging in. The will include different sections such as:

- Adding guests – add guest information to database
- Viewing and updating guest information
- Deleting guest (only available for managers and administrators)
- Guest search – find guest using there name or GuestID
- Validation – check if the guest information is valid such as a name being a string

Field name	Description	Data type
GuestID	Unique identifier for each guest	Integer
FirstName	Stores guests first name	String
LastName	Stores guests last name	String
PhoneNumber	Stores guests phone number	String
Email address	Stores guests email address	String
Home address	Stores guests home address	String

Room management:

After logging in the user will have access to the room management screen, although this will be mostly needed for reception so it will be more accessible to them. It will show each block of the hotel and when clicked on will show each room on the block. There will be some different sized rooms and there will be different prices based on that. There will also be a premium block which will have more expensive rooms and some that have pools outside. After discussing details with the guest, the receptionist will assign a guest to a roomID based on their requests.

ISPO chart:

Feature	Input	Process	Storage	Output
Login	Email, password	Check password, check role	Staff table	Role based menu or error message
Bookings	Guest, room, check in, check out	Check availability, Lock room, calculate cost, save booking	Booking and room tables	Booking confirmation, updated availability
Edit/cancel booking	Booking ID, new details or cancel request	Re-check availability, recalculate cost and update status	Booking table	Updated booking or cancellation confirmation
Guest records	Guest details or search term	Validate fields, encrypt sensitive data, search the records	Guest table	Guest profile or search results

Payments	BookingID, amount, method	Calculate balance, apply any approved discounts	Booking and payment table	On screen report
Reports	Report type, date range	Calculate occupancy and revenue	Booking and payments table	On screen report

Description of data structures:

Object oriented programming will be used. This program will have 3 main classes:

1. Booking – stores all details of a single booking

- a) Attributes
- BookingID
- Guest ID
- RoomNumber
- CheckInDate
- CheckOutDate
- TotalCost

2. Guest – stores personal details of all guests

- a) Attributes
- GuestID
- FullName
- Address
- Email
- PhoneNumber
- IDType
- IDNumber
- DateOfBirth
- CardDetails

3. Room – Stores the details of each of the 160 rooms

- a) attributes
- RoomNumber
- Block
- Floor

- RoomType
- PricePerNight
- Availability status